CHEMISTRY ADAPTLY

ALLOTROPES OF CARBON

KEY VOCABULARY

Allotrope: different structural forms of the same element.

Graphene: a single layer of graphite — one atom thick.

Fullerene: carbon molecules shaped like balls or tubes.

Nanotube: a cylinder of carbon atoms — strong and conductive.

COMMON MISCONCEPTIONS

~~Diamond and graphite are different elements.~~

✓ Both are pure carbon — just arranged differently.

~~Fullerenes are giant covalent.~~

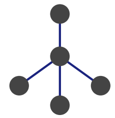
✓ Fullerenes are molecules, not giant lattices.

~~Graphene is the same as graphite.~~

✓ Graphene is just one single layer of graphite.

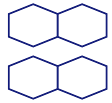
FORMS OF CARBON

DIAMOND



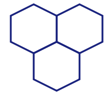
4 bonds each — hard

GRAPHITE



3 bonds — layers slide

GRAPHENE



single layer — strong

1 DEFINE ALLOTROPE

State what is meant by an allotrope. **[1 mark]**

2 GRAPHENE

Describe the structure of graphene. **[2 marks]**

3 WHY CONDUCTIVE

Explain why graphene and nanotubes conduct electricity. **[2 marks]**

4 FULLERENES

Suggest one use of fullerenes and why their structure suits it. **[2 marks]**

5 FILL THE GAPS

Complete the sentence using the word bank. **[2 marks]**

Graphene is a _____ layer of carbon where each atom bonds to _____ others.

one	single	three	four

6 STRENGTH

Suggest why carbon nanotubes are used to strengthen materials. **[2 marks]**

TIP: *Same element, different structure → different properties.*

✓ **Think: Why does each form of carbon have a free electron except diamond?**